CI interpretation/reporting + method-selection tree

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Need to choose among Z, t, proportion, FPC or bootstrap.
- Asked to interpret/report a finished interval.
- Closest competitor depends on parameter, σ status, normality and finite sampling.

2 DECISION / METHOD

Tree:
1. Parameter: mean or proportion?
2. Mean: σ known → Z; σ unknown → t.
3. Proportion → Z CI using p.
4. If approximate normality is doubtful / the population is clearly non-normal, use bootstrap where the course/question supports it; small n makes the normality issue especially important.
5. Finite without replacement and n/N>.05 → multiply SE by FPC.

3 FORMULA

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report: estimate, lower, upper,
confidence level, n, units/context
CI describes parameter uncertainty,
not individual observations
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4 FORMULA SHEET VS WHAT I KNOW

Gives: method-specific SEs and sample-size formulas.
Still know: selection tree and correct repeated-sampling interpretation.

5 EXCEL ROUTE

Use the chosen route:
- Z: NORM.S.INV
- t: T.INV.2T, STDEV.S, df=n-1
- proportion: SQRT(p*(1-p)/n)
- FPC: SQRT((N-n)/(N-1))
- bootstrap: resample and empirical cutoffs.

6 TINY WORKED EXAMPLE

A 95% CI for μ is (12.4,15.1). Say: 'We are 95% confident the population mean lies between 12.4 and 15.1 units.'

7 TOP EXAM TRAPS

- '95% of data lie in the CI.'
- 'There is a 95% probability fixed μ is in this computed interval.'
- Using t for a proportion.
- Forgetting FPC/normality checks.

8 SAY ALOUD

Choose by parameter and σ status, then interpret the interval as uncertainty about the population parameter.

9 EXIT CHECK

Unknown σ, small normal sample, mean parameter: which distribution and df?

ONE-GLANCE VISUAL ROUTE

